



Experimental Investigation of a Variable Collective Pitch Propeller in Near-Edgewise Flight

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We extend our experiments on fixed and variable collective pitch propellers in forward flight, to near-edgewise flight conditions. Different combinations of blade pitch, incidence angle, and flight speed were tested in an open-jet test section wind tunnel, at incidence angles of 75 to 100 degrees. Higher thrust and power consumption are found at higher incidence angles, which agrees with the previous study and literature. Power required at constant vertical thrust is compared across different incidence angles and advance ratios, to evaluate the efficiency of the variable collective pitch propeller, and to assess whether a rotor-borne vehicle should tilt forward in forward flight or instead remain in purely edgewise configuration and use separate horizontal thrusters for forward propulsion.

I. Nomenclature

C_T	= Propeller Thrust Coefficient; $C_T = \frac{T}{\rho n^2 D^4}$	J	= Propeller Advance Ratio; $J = \frac{V_\infty}{nD}$
C_{TA}	= Propeller Axial Thrust Coefficient	n	= Propeller Rotational Speed, (RPS)
C_{TIP}	= Propeller In-Plane Thrust Coefficient	P	= Power, (W)
C_{TH}	= Propeller Horizontal Thrust Coefficient	γ	= Propeller Pitch, (m)
C_{TV}	= Propeller Vertical Thrust Coefficient	Q	= Torque, (Nm)
C_P	= Power Coefficient; $C_P = 2\pi C_Q = \frac{P}{\rho n^3 D^5}$	R	= Propeller Radius, (m)
C_Q	= Propeller Torque Coefficient; $C_Q = \frac{Q}{\rho n^2 D^5}$	T	= Thrust, (N)
C_m	= Propeller Pitching Moment Coefficient	V_∞	= Freestream Velocity, (m/s)
C_n	= Propeller Yawing Moment Coefficient	ρ	= Air Density, (kg/m ³)
D	= Propeller Diameter, (m)	δ	= Propeller Blade Pitch Angle, (deg)
		η	= Propeller Efficiency, $\eta = \frac{C_T J}{C_P}$
		θ	= Propeller-plane Incidence Angle, (deg)

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II. Introduction

The increasing interest in Advanced Air Mobility (AAM) broadens the usage of propeller-driven aerial vehicles, including unmanned aerial vehicles (UAVs) and electric Vertical take-off and Landing (eVTOL) air taxis. This expands the application scenario of the classical propeller from conventional fixed-wing aircraft to various forward flight conditions where the propeller disk is placed under different incidence angles in near-edgewise flight, similar to the application of a helicopter rotor. The mainstream concept of a multi-copter, whether strictly rotor-borne or assisted by conventional wings, is predicated on some plurality of fixed-pitch propellers. The propeller rotational speed varies for flight control and for prosecuting the various performance tasks (climb, forward/backward acceleration, steady level cruise, and so on). These aircraft generally tilt forward to fly forward, so that the lifting propellers – that is, the only propellers – are now at some incidence angle away from purely edgewise flight.

The classical propeller experiment is for fixed-wing aircraft in forward flight, where the propeller sees an incidence angle $\theta \sim 0^\circ$, which is to say, faces the oncoming stream head-on [1-7]. Assuming fixed pitch, and pitch values not at or above stall in static conditions, at lower advance ratios ($J < \sim 0.2$), the thrust coefficient (C_T) and power coefficient (C_P) remain relatively constant. As the advance ratio increases beyond this, both C_T and C_P fall as the propeller unloads and eventually reaches zero at a sufficiently high J , depending on the propeller pitch (γ). The value of zero-thrust advance ratio can, of course, be delayed with higher collective pitch. The effect of doing so, for variable collective-pitch drone propellers, was considered in our prior work [15] at incidence angles up to $\theta \sim 20^\circ$.

Propellers at higher incidence angles (non-axial flight) have also been investigated [4,5,8,9-11,14]. McCormick [6] notes that classical momentum theory is viable for both zero incidence (purely axial) and 90-degree incidence (purely edgewise) flight, concluding that if the approximation works for these two extremes, it should be applicable to all intermediate cases. Simmons and Hatke [4], McLemore and Cannon [5], Ribner [8], and Cai and Gunasekaran [14] investigated fixed-pitch propellers under a steady freestream at incidence angles between 0 and 90 degrees. The axial thrust coefficient (C_{TA}) and axial torque coefficient (C_{QA}) decrease with the increasing advance ratio at small θ , as expected. However, the C_{TA} and C_{QA} vs. J curve plateaus at higher incidence angles. C_{TA} and C_{QA} (Figure 1) are nearly independent of J at $\theta \sim 60^\circ$, and show an increase with J at higher θ and approaching edgewise flight ($\theta = 90^\circ$). In other words, the propeller's ability to generate axial thrust increases when operating in edgewise flight. This, of course, is the standard helicopter theory result. But for fixed pitch propellers, due to the asymmetric blade loading at higher θ , higher pitching moment (C_m) and yawing moment (C_n) are observed. A good correlation is found for these moments vs. incidence angle, when plotted vs. the normal component of the advance ratio [4,11,14].

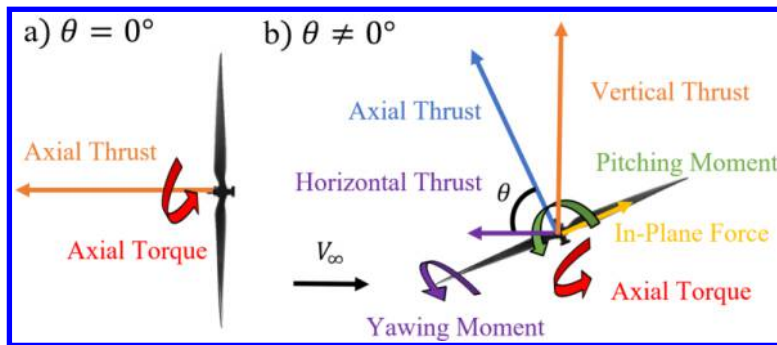


Figure 1. Propeller Force and Moment Definition at a) $\theta = 0^\circ$ and b) $\theta \neq 0^\circ$

The study of a variable collective pitch propeller at nonzero incidence is more limited, as it neither fits the context of helicopters, nor that of multi-copter drones. Yaggy and Rogallo [11] tested a three-bladed rotor with variable pitch for $0^\circ \leq \theta \leq 85^\circ$. Besides the aforementioned findings, a significant difference in the rotor yawing moment (C_n) is found at the same θ under the same incidence angle and different blade pitch angles (δ), while a smaller difference between the rotor pitching moment (C_m) is observed under the same conditions at different δ . Our previous study [15] investigated the forward flight performance of the same variable pitch propeller used in the current study at near-axial flight conditions ($-10^\circ \leq \theta \leq 20^\circ$). Using the propeller performance between C_T and δ in hover and the forward axial flight performance at $\delta = 0^\circ$, good thrust predictability of the variable pitch propeller is found in near-axial forward flight.

Cai [16] investigated forward thrust and vertical lift generation (Figure 1) for fixed-pitch propellers with different blade geometries, at various incidence angles. At a small incidence angle, a propeller with a lower blade pitch distribution showed higher efficiency in axial and forward thrust generation at lower forward advance ratios; and vice versa at higher forward advance ratios [1-3, 11, 16]. This trend is preserved when increasing the propeller incidence angle until reaching nearly-edgewise flight, where blade stall occurs on the propeller with a higher pitch-to-diameter ratio, especially at higher advance ratios [11,12]. Therefore, a propeller with a lower collective pitch may be more favorable for high-speed edgewise flight applications. Implementation of a variable pitch propeller potentially enables the propeller disk to operate at higher efficiency across the incidence angle range, from axial flight to edgewise flight ($0^\circ < \theta < 90^\circ$), and across the range of advance ratios.

The present study extends Yaggy and Rogallo [11] to Reynolds numbers, parameters, and components relevant to modern small drones, measuring all six components of force and moment. We consider collective pitch from the factory setting implied for a fixed-pitch propeller ($\delta = 0^\circ$) to past blade stall at a given advance ratio. We then down-scope back to fixed pitch but consider what happens to the power required to produce a given vertical thrust, in a range of near-edgewise incidence angles and advance ratios. Besides building a propeller performance database, we ask a question relevant to drone configurations: should a rotor-borne drone fly forward by tilting its body axis, thus getting a horizontal component of lifting-rotor thrust into the flight direction? This is how most drones fly today. Or should there be a separate set of horizontal thrusters or propellers, whose job it is, to provide thrust in the horizontal direction, while the drone itself does not tilt, and the lifting rotors/propellers remain in purely edgewise flight? The matter depends on the total power in forward flight: fuselage drag (omitted in the current study) and lifting rotor drag, both multiplied by the flight speed; and the lifting rotor induced power, all for a given drone weight (that is, vertical component of lifter thrust). What happens to lifting rotor induced power, and the force in the flight direction – that is, the “drag force on the rotor” – as a function of angle deviating from purely edgewise flight? This point is subsumed by helicopter theory, but in a way that isn’t immediately clear for drones. The helicopter rotor has induced power and profile power... theoretically separate quantities, but they are of course only measurable together. A helicopter rotor tilts forward, perhaps with cyclic pitch, to overcome rotor drag. There is a total power consumed; in doing so... and this total power is plotted as a function of flight speed. But if there is no cyclic pitch – and there isn’t, in the drone community – then we are confronted with the question, of how much power (or literal tilt of the rotor axis) is needed to fly forward, and how much, separately, is needed to overcome vehicle weight, and how all of this varies with flight speed, or nondimensionally, with advance ratio.

III. Experimental Setup

A) Data Acquisition

Experiments were conducted at the University of Dayton Low-Speed Wind Tunnel (UD-LSWT) in the open jet test section. The tunnel has an inlet contraction ratio of 16:1, a 762 mm x 762 mm testing section inlet, and a 1.37 m x 1.37 m collector. The turbulence intensity is 0.1%, measured by a hot wire anemometer.

A sketch of the experimental setup is shown in Figure 2. For the variable pitch propeller assembly, three blades taken from T-motor FA 16.2x5.3 propellers are paired with a T-motor V505 KV 260 brushless motor and a T-motor AT 115A 14S electronic speed controller [19]. Each blade is retained in a bearing cage that reacts against centripetal force in an essentially rigid coupling. Variable collective pitch is accomplished via a single servo that drives a central pushrod through the motor, to a common linkage mechanism in the bearing-cage, and finally, a pitch-link at each blade grip. Conceptually, the approach resembles a helicopter tail rotor assembly. There is no cyclic pitch variation. Blade collective pitch is statically calibrated to reference measured blade angle to servo pulse width modulation (PWM) commands.

For the fixed pitch propeller, the two-bladed KDE 12.5x4.3 propeller is paired with the KDE2814XF-515 brushless motor [20] and a T-motor AT 115A 14S electronic speed controller. Both motors are powered by a GW Instek PSW 30-108, 1080W power supply.

For the variable pitch propeller, the propeller and motor assembly are attached to an ATI Industrial Automation Gamma six-component force/torque transducer [21]. The transducer has a maximum of 65N on its x and y-axis, 200N on its z-axis, and a maximum torque of 5 Nm for all axes. For the fixed-pitch propeller, due to a lower magnitude of thrust and torque produced, the ATI Industrial Automation Mini-40 six-component force/torque transducer was used. Thrust is the axial force along the z-axis of the sensor. The motor torque is the moment about the z-axis. This produces the classical torque coefficient, which, when multiplied by rotational speed, gives the power coefficient, as usual. The positive y-axis of the sensor is also defined in Figure 2 and the orientation follows the right-hand rule for the positive x-axis.

The transducer is mounted on a 3D-printed Polycarbonate interface, bolted to an 80-20 aluminum extrusion frame. The propeller can be tilted with respect to the tunnel freestream by changing the configuration of the aluminum frame. The entire assembly is attached to a 1.22 x 0.92 x 2.44-meter external aluminum frame to improve the rigidity of the testing stand and minimize vibration. The external aluminum frame is secured to the lab floor, the inlet of the wind tunnel, and the ceiling of the wind tunnel plenum. A 10-mW laser and one Thorlabs PDA10A Photodiode are used to measure the propeller rotational speed. A picture of the experimental setup is shown in Figure 3.

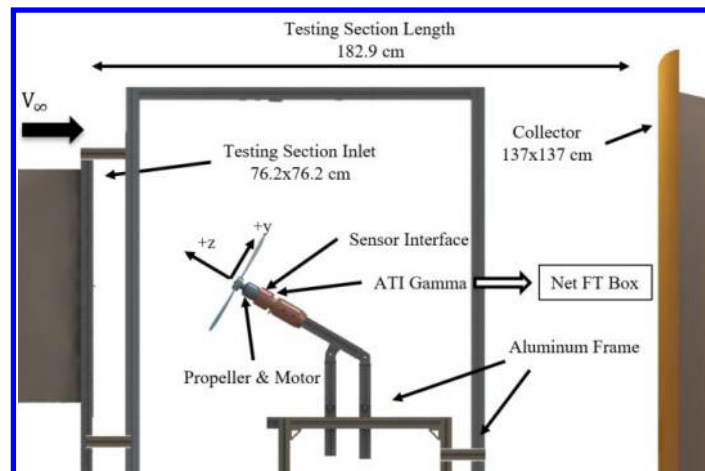


Figure 2. Schematic of the Experimental Setup at the UD-LSWT.

Data was sampled at the rate of 1000 Hz for a 10-second duration at each data point. Two trials were run in each case to improve repeatability. To minimize sensor drift, tare values with the propeller powered off were taken before and after each test. The average of the two tares was subtracted from the respective thrust and torque readings. If the difference between the before and after tare value was greater than 0.2N of force on any axis, the experimental data was rejected. A Finite Impulse Response (FIR) band-stop filter [22] was applied to the data to filter out the vibration frequency caused by the propeller rotation and the natural frequency of the aluminum test frame. Tap tests on the test stand with the propeller powered off showed a structural natural frequency of ~ 57 Hz and a harmonic frequency at ~ 114 Hz. With due to the uneven blade loading in near-edgewise flight conditions, significant vibration is experienced by the propeller and the force balance. Any dataset that shows abnormally high random uncertainty was rejected during the data post-processing.

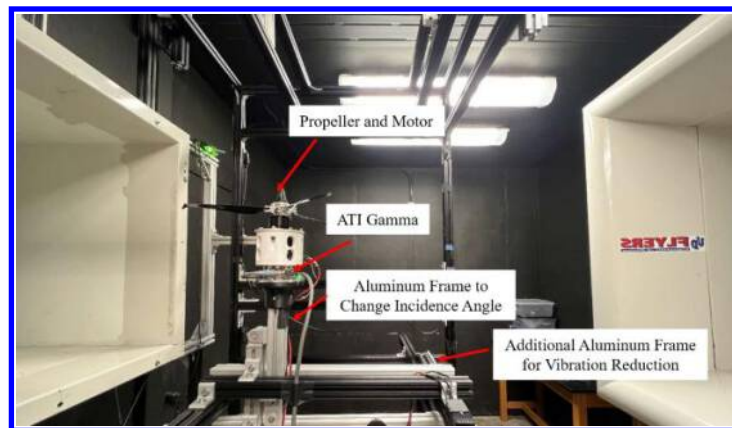


Figure 3. Experimental setup at the UD-LSWT.

B) Test Matrix

The test matrix is shown in Table 1. The variable pitch propeller rotational speed varies between 40 and 80 RPS under different collective pitch angles (δ). The Reynolds number is calculated at the 75% blade radius. The freestream velocity varies from 0 m/s to 30 m/s, resulting in an advance ratio, J , between 0 and 1.25. Three incidence angles (θ) near-edgewise flight is tested. The blade collective pitch angle, measured based on the angle of the flat portion of each propeller blade at its grip, varied from $+1.0$ to $+20.8$ degrees. $\delta = 0^\circ$ is the ‘factory default’. A positive blade pitch angle indicates the effective blade pitch distribution is higher than the ‘factory default’. The corresponding propeller root blade pitch angle vs. servo input PWM is nearly linear, with a scatter of less than 0.5 degrees between different trials [15].

For the fixed-pitch propeller, the rotational speed is kept near 84 RPS. The wind tunnel freestream velocity varies from 0 m/s to 25 m/s, resulting in an advance ratio of $0 \leq J \leq 1.0$. The incidence angle (θ) varies from 70 to 100 degrees, in 5-degree increments. At $\theta > 90^\circ$, the propeller is tilted past the pure edgewise condition and resembles the ‘braking condition’, or perhaps an autogiro.

Table 1. Test Matrix

Propeller	Rotational Speed (RPS)	Peak Reynolds Number (x1000)	Advance Ratio	Incidence Angle (θ) (Deg)	Collective Pitch Angle (δ) (Deg)
T-motor FA 16.2x5.3	40 - 80	190	0 – 1.25	75, 82.5, 90	1.0 - 20.8
KDE 12.5x4.3	83 - 86	102	0 – 1.00	70, 75, 80, 85, 90, 95, 100	N/A

IV. Results

A. Force Measurement of the Variable Collective Pitch Propeller

The force balance (Figure 2) directly measures the axial and in-plane force (Figure 1.b), and this is converted to axial thrust coefficient (C_{TA}) and in-plane thrust coefficient (C_{TIP}) shown in Figure 4 and Figure 5. The legend shows δ from 1.0 to 20.8 degrees above the propeller manufacturer's factory zero-angle ($\delta = 0^\circ$). In general, C_{TA} increases with δ for all cases, as expected, because the propeller is more heavily loaded. At $\theta = 75^\circ$, C_{TA} remains almost constant with J across all δ tested. This agrees with the trend observed in our previous study, and the literature [11, 14, 17, 18]. Larger θ results in larger C_{TA} at higher J , with $\theta = 90^\circ$ yielding the highest slope of C_{TA} vs. J , as evidently the propeller is processing more momentum-flux.

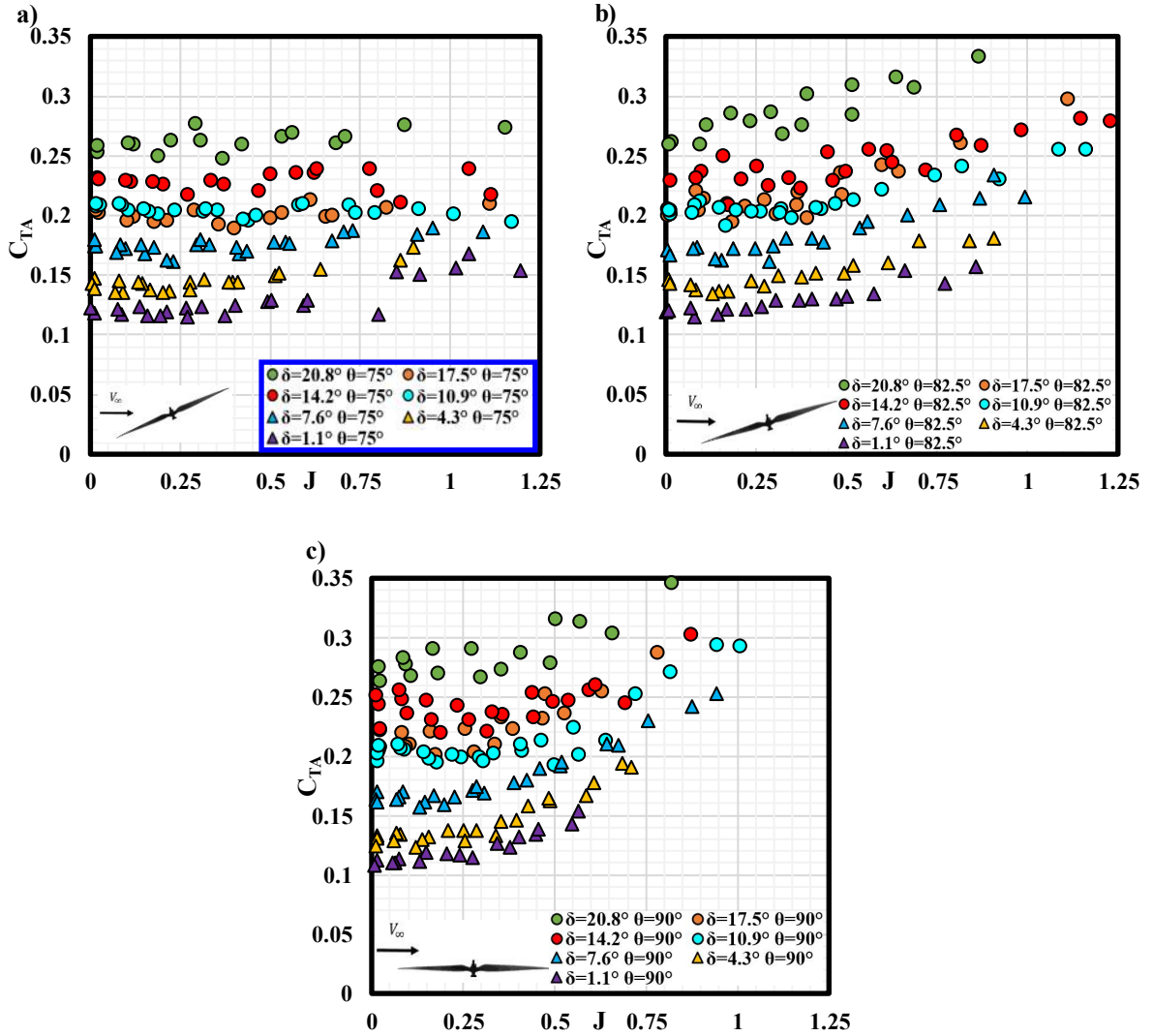


Figure 4. C_{TA} vs. J at various δ at a) $\theta = 75^\circ$, b) $\theta = 82.5^\circ$ and c) $\theta = 90^\circ$.

C_{TIP} is largely independent of δ , increasing more or less linearly with J . The slopes of C_{TIP} vs. J are similar for $\theta = 75^\circ$ and 82.5° while a smaller slope is observed at $\theta = 90^\circ$, indicating a smaller drag penalty for the edgewise flight propeller, especially at higher advanced ratios.

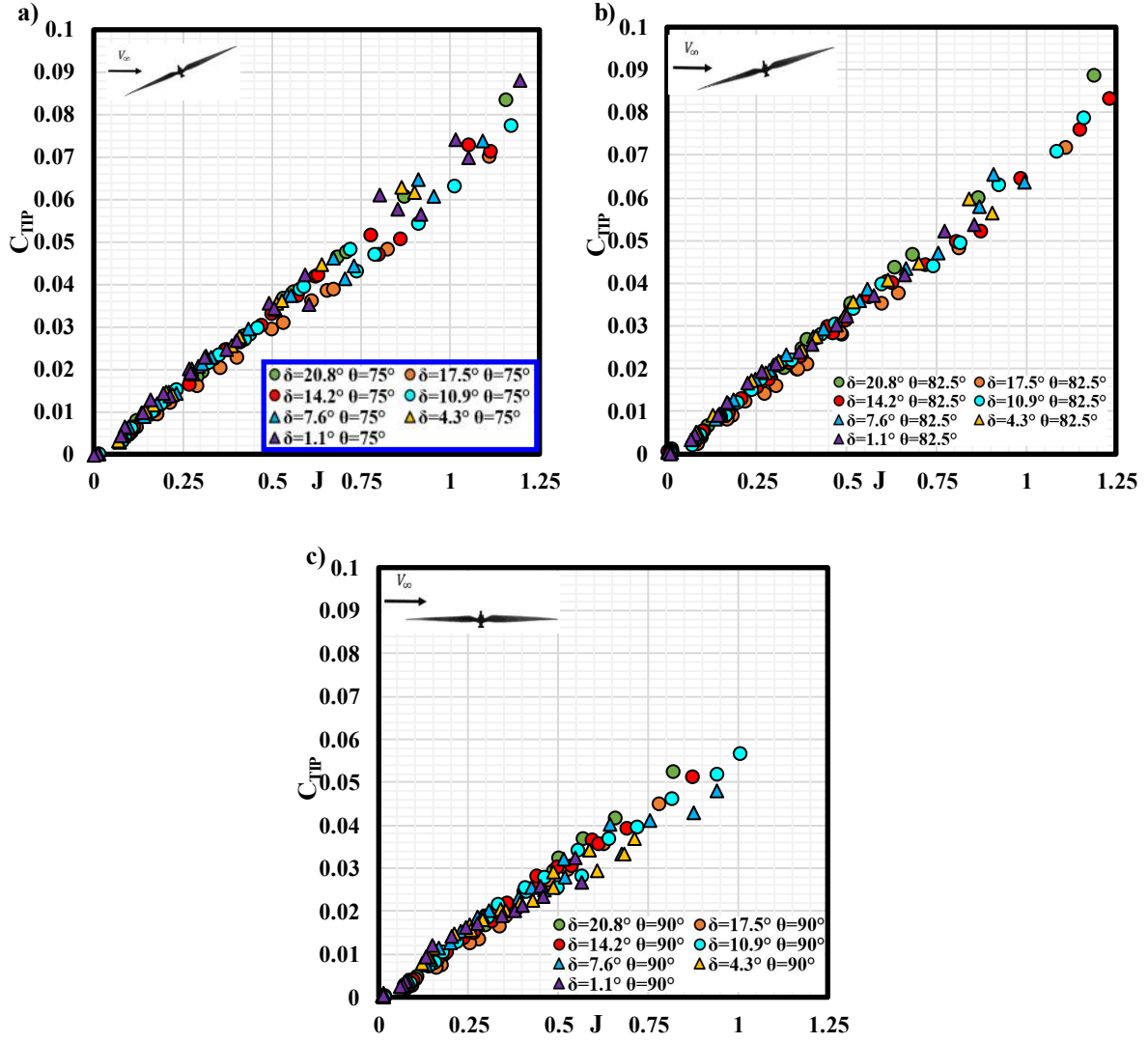


Figure 5. C_{TIP} vs. J at various δ at a) $\theta = 75^\circ$, b) $\theta = 82.5^\circ$ and c) $\theta = 90^\circ$.

The axial and in-plane forces are then converted to the horizontal thrust coefficient (C_{TH}) and vertical thrust coefficient (C_{TV}), in the rubric of Figure 1.b. A negative value of C_{TH} means a drag experienced by the propeller. Figure 6 shows C_{TV} vs. J vs. δ at the three θ cases tested. A higher C_{TV} is measured at a higher δ under the same J , following the trend of C_{TA} . With increasing forward flight speed, C_{TV} also increases. Overall, higher θ , meaning going closer to fully edgewise flight, results in a higher C_{TV} from higher axial thrust (C_{TA}) measured on the propeller, and from trigonometry.

Figure 7 shows the horizontal thrust coefficient (C_{TH}) for the cases in Figure 6. Higher C_{TH} is measured at a higher δ for a given J and θ , because of higher axial thrust. As J increases, C_{TH} falls nearly linearly, as a result of the higher in-plane force, and thus a higher drag force on the propeller. It is worth noting that the slope of $\frac{dC_{TH}}{dJ}$ remains almost constant at all times θ and δ tested, with an average magnitude of -0.0592 .

Eventually, C_{TH} becomes negative, meaning no more forward net thrust-available at this or higher flight speeds. If θ is increased, C_{TH} falls as a result of a combination of propeller in-plane force and a lower forward thrust resolved from the axial thrust through the incidence angle. As θ is increased, the spread in C_{TH} with δ

decreases. C_{TH} collapses at $\theta = 90^\circ$, as the propeller in-plane force (Figure 5) is the only force acting in the horizontal direction.

The corresponding power coefficient (C_P), calculated based on the torque at the propeller shaft, for the cases in Figure 4 through Figure 7 is shown in Figure 8. The trend follows those seen in the literature and our previous studies [11, 14, 16], where a higher δ results in a higher power consumption at the same J .

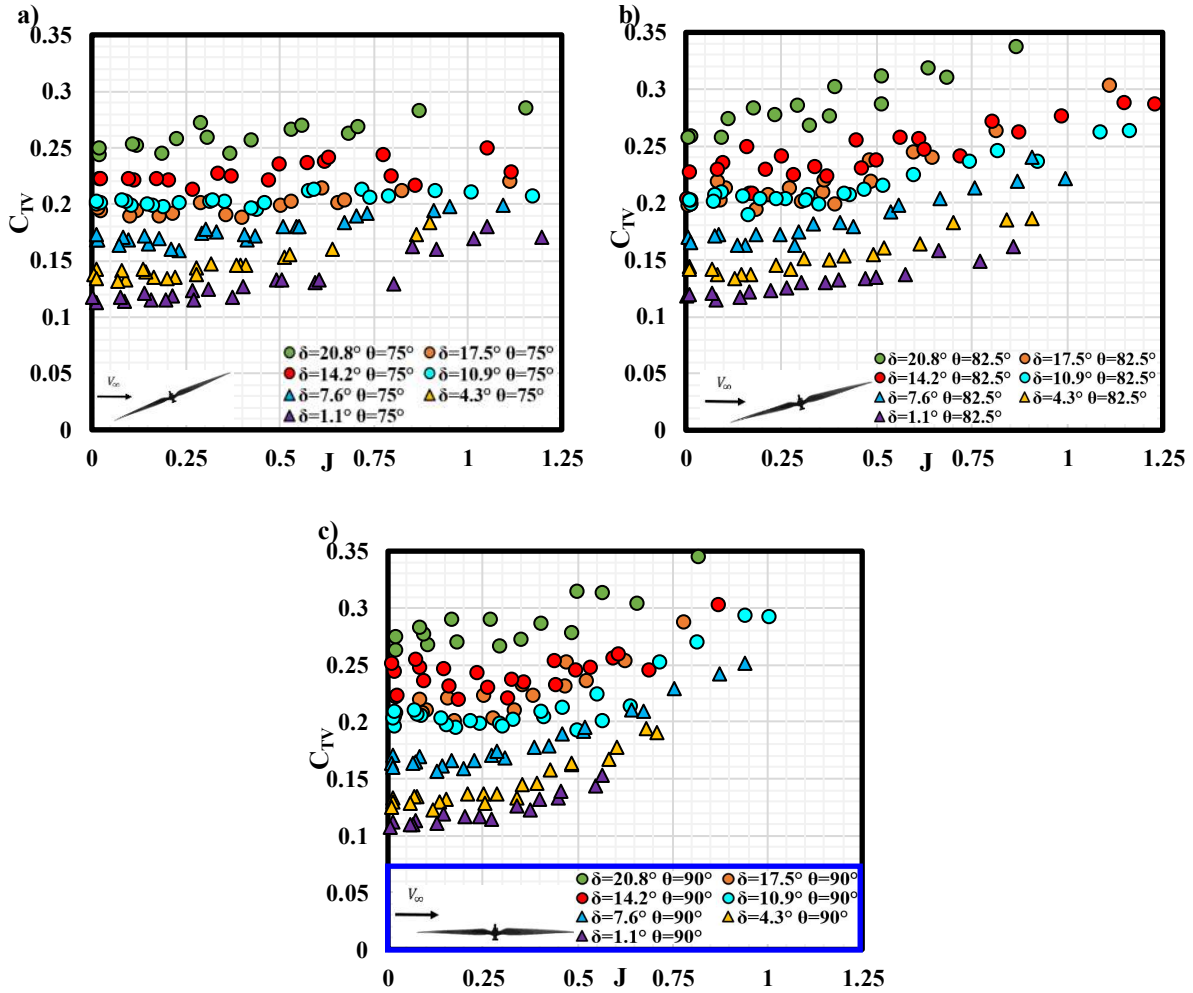
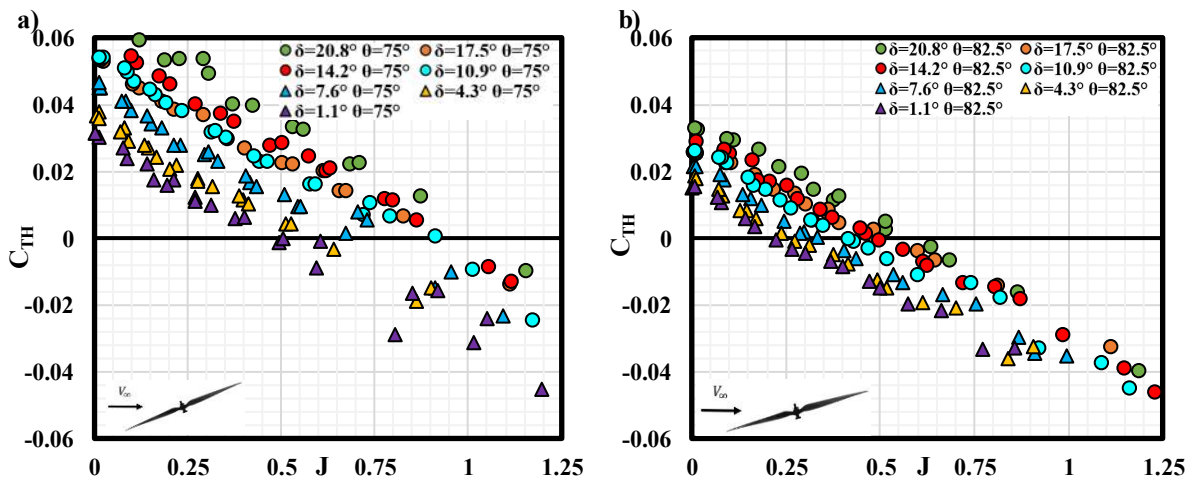


Figure 6. C_{TV} vs. J at various δ at a) $\theta = 75^\circ$, b) $\theta = 82.5^\circ$ and c) $\theta = 90^\circ$.



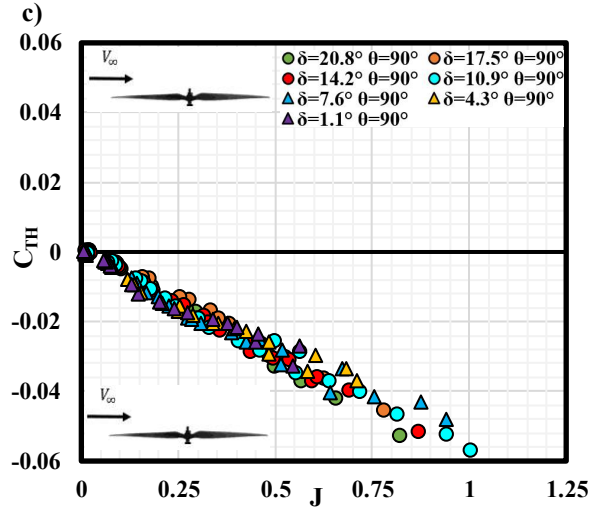


Figure 7. C_{TH} vs. J at various δ at a) $\theta = 75^\circ$, b) $\theta = 82.5^\circ$ and c) $\theta = 90^\circ$.

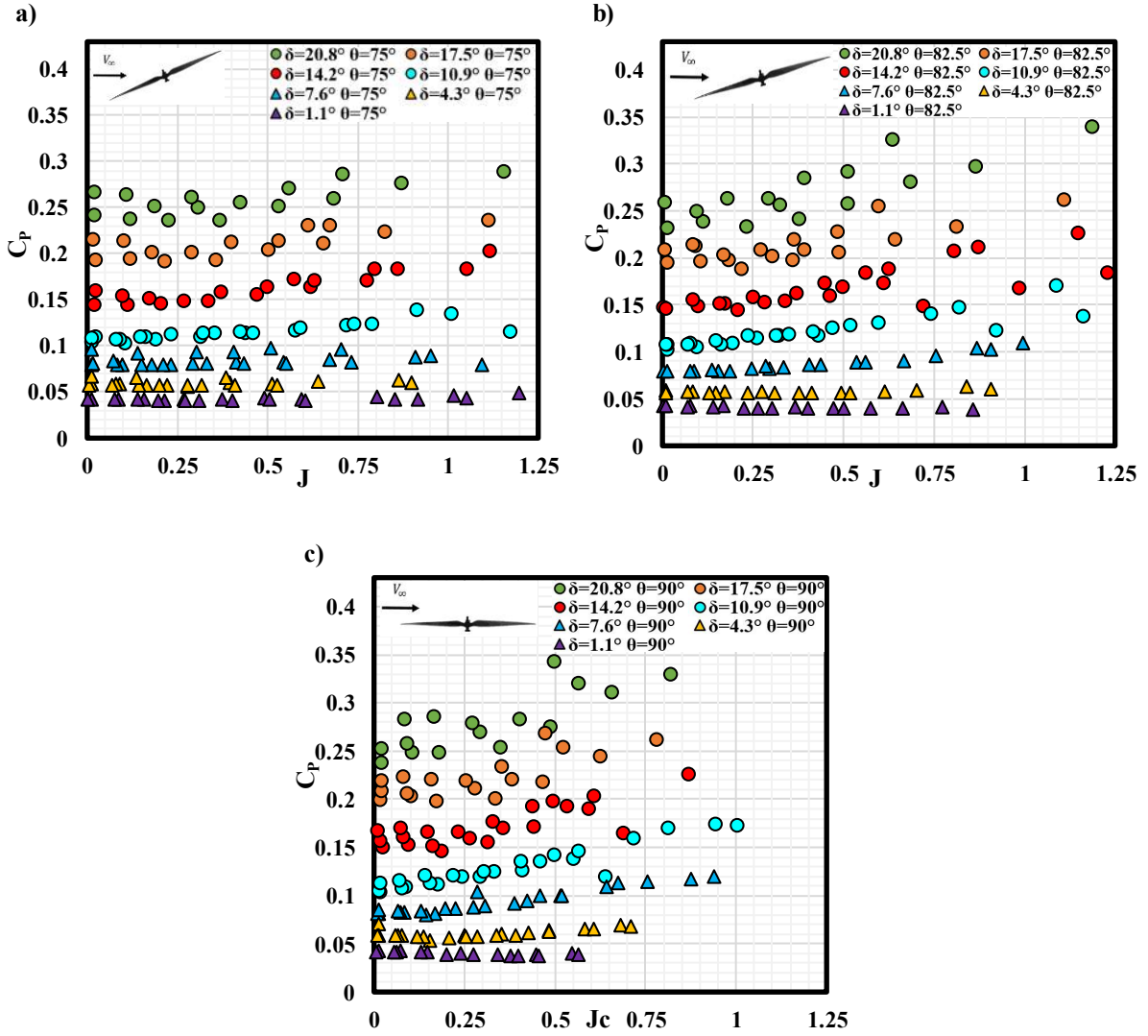


Figure 8. C_P vs. J at various δ at a) $\theta = 75^\circ$, b) $\theta = 82.5^\circ$ and c) $\theta = 90^\circ$.

B. Power Consumption Analysis of the Variable Collective Pitch Propeller

For propellers in axial flight, efficiency is the ratio of the useful work done by producing forward thrust, to the shaft power consumed. As the propeller incidence angle is increased, some of the axial thrust contributes to thrust in the vertical direction. Therefore, the traditional propeller efficiency might not be applicable to edgewise flight conditions. Results for the traditional propeller efficiency calculated by $\eta_A = C_{TA}J/C_p$ at $\theta = 75^\circ$ are shown in Figure 9a. A higher η_A is observed at a higher δ at the same J . η_A also increases with J and eventually reaches a spurious value larger than 1.0. This is a result of a higher C_{TA} at higher J at large incidence angles, as seen in Figure 4, which leads to a much higher product of $C_{TA}J$ for a high advance ratio. Alternatively, we consider that the useful work is only done by the propeller horizontal thrust, and calculate the propeller efficiency in terms of the horizontal thrust as:

$$\eta_H = \frac{C_{TH}J}{CP} \quad (1)$$

The resulting η_H is shown in Figure 9b, where η_H increases with J and reaches its maximum, then eventually falls to 0 at a higher J , as is the expected case for propeller axial flight. Beyond this J , the propeller loses its ability to generate horizontal forward thrust, as seen in Figure 7. When increasing δ , a higher η_H is observed at the same J . Such a trend follows the traditional variable collective pitch propeller efficiency in axial flight, as discussed in our previous study [15]. This horizontal-only presentation suggests that higher collective pitch improves the efficiency, with increasing advance ratio, of a propeller inclined forward from edgewise flight. But it is deceptive, because it ignores the vertical thrust component, which is necessary to oppose the presumptive weight of the flight vehicle.

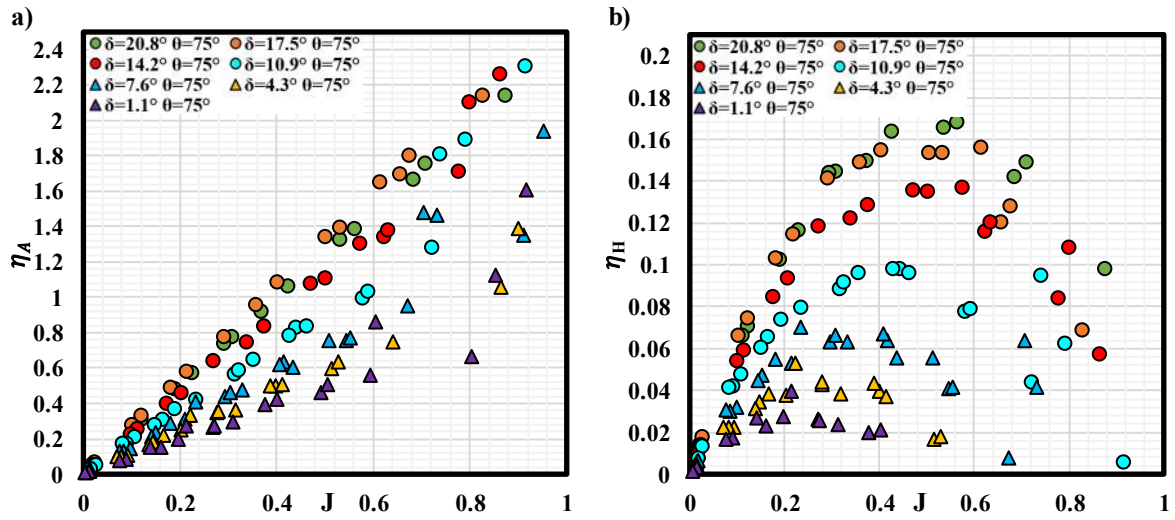


Figure 9. a) η_A vs. J and b) η_H vs. J at various δ and $\theta = 75^\circ$.

On the other hand, in hover at $\theta = 90^\circ$ and $J = 0$, the only power consumed by the propeller is the induced power. The horizontal component of thrust (or drag) is irrelevant. Rotor efficiency is represented as the Figure of Merit [13], which is the ratio of the ideal induced aerodynamic power to the actual propeller shaft power. The Figure of Merit is only really applicable in hover. When operating at edgewise or nearly-edgewise flight ($\theta \sim 90^\circ$), the total power required contains the induced power necessary to produce the vertical thrust, and the power

required to produce the horizontal thrust. Therefore, neither propeller efficiency in axial flight nor the Figure of Merit would be a good representation of the overall propeller efficiency.

The traditional helicopter approach [13] is to plot dimensional power – necessary to produce a dimensional vertical thrust – vs. dimensional flight speed. This illustrates the usual helicopter or VTOL power curve, with a peak at hover, zero slope of power vs. speed just away from hover, then a decline towards some forward speed at which power-required is minimal, and finally a rise, more or less as a cubic, as flight speed increases further. Applied to the present case, there are several difficulties. First, such a presentation necessarily includes fuselage drag, and while this is useful in full flight vehicle system analysis, it clouds the issue of what is happening to a single propeller in isolation. Second, dimensional results are a useful illustration for a typical helicopter, but for work on a wind-tunnel-scale propeller, we would like to nondimensionalize forces to coefficients and speeds to advance ratios. As we shall see in the following, such nondimensionalization is problematic for near-edgewise flight. We already saw one such problem in Figure 9.

One representation of efficiency is the ratio of the power coefficient (C_P) to the axial thrust coefficient (C_{TA}), which represents the propeller power consumption per unit thrust generated. Results for the variable collective pitch propeller are shown in Figure 10. Across all θ cases tested, a higher ratio of C_P/C_{TA} is measured at a higher δ , as a result of a higher disk loading, indicating a lower overall efficiency. At lower δ , a small reduction of C_P/C_{TA} is observed with increasing J , especially at $\theta = 90^\circ$. This trend is reversed at higher δ . Both pieces of evidence show that a higher collective pitch does not benefit the efficiency of propeller power consumption in axial thrust generation. Meanwhile, at the same δ and J , a higher incidence angle (θ) results in a lower C_P/C_{TA} indicating a higher efficiency in generating axial thrust, as a result of the increment of C_{TA} and a decrement of C_P under the same conditions, as seen in Figure 4 and Figure 8. This, of course, is the usual result in helicopter theory, which states that total power consumption is lower at some forward flight speed than in hover.

C. Propeller Power Consumption in Near-Edgewise Flight

As seen in the previous section, unlike for axial flight, increasing the propeller collective pitch in near-edgewise flight results in a higher power consumption when generating a constant vertical thrust. Operating the collective blade pitch angle at the factory default condition ($\delta = 0^\circ$) produces the best overall performance in edgewise flight. Therefore, we turn to a fixed-pitch propeller in what follows, to answer the questions: at what incidence angle does the drone consume the least amount of power, or be most efficient, near-edgewise flight condition?

Using the KDE 12.5x4.3 propeller following our previous study [17,18], the axial and in-plane thrust coefficients in our current study are plotted along with our previous study, in Figure 11, showing good agreement. C_{TA} continues to increase at $\theta > 90^\circ$ through the maximum incidence angle tested ($\theta = 100^\circ$), where effectively the propeller resembles the orientation of an autogiro rotor. C_{TA} will eventually reach its maximum at some J at a high enough incidence angle [4]. Meanwhile, the spread of C_{TIP} between different θ is smaller. C_{TN} increases with J , as in Figure 5. When increasing θ at the same J , C_{TN} increases and reaches its maximum near $\theta = 70^\circ$. Further increase of θ results in the decrease of C_{TIP} at the same J , where the lowest C_{TN} magnitude occurs at $\theta = 100^\circ$. This indicates that the maximum propeller in-plane force, which contributes to the drag force on the horizontal axis in forward flight, does not occur under pure edgewise flight conditions, but at a lower incidence angle.

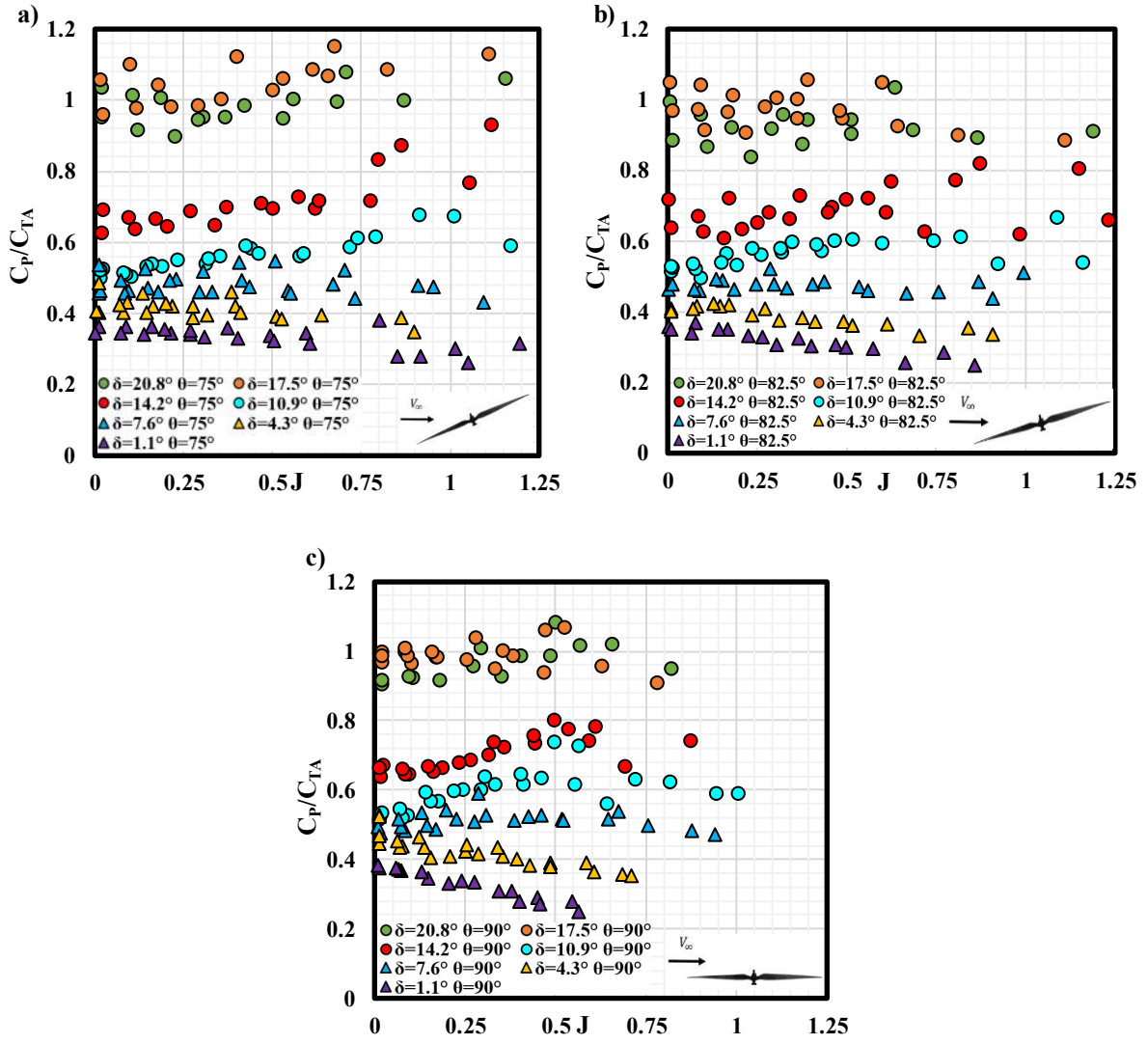


Figure 10. C_p/C_{TA} vs. J at various δ at a) $\theta = 75^\circ$, b) $\theta = 82.5^\circ$ and c) $\theta = 90^\circ$.

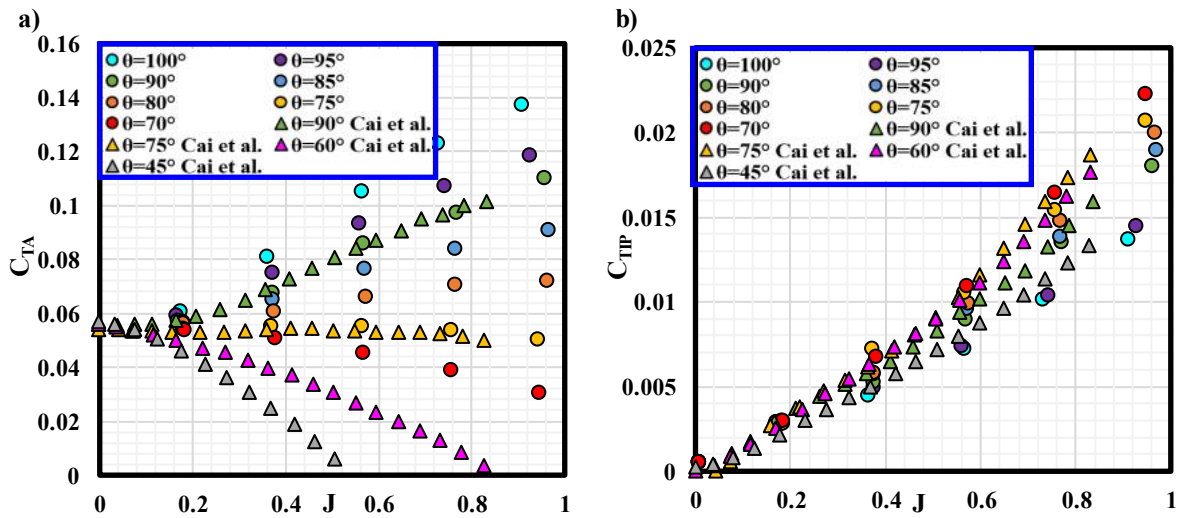


Figure 11. a) C_{TA} vs. J and b) C_{TIP} vs. J for fixed fixed-pitch propeller at various θ , and previous results [17,18].

Using the same process as for Figure 6 and Figure 7, the horizontal and vertical thrust coefficients are shown in Figure 12. The overall trends are also similar to those in Figure 6 and Figure 7, with a higher θ resulting in a higher C_{TV} and lower C_{TH} at the same J . Similar to the results in Figure 7, C_{TH} varies almost linearly with J . Slope varies slightly with incidence angle, but averages to ~ -0.025 . The significance of this slope – that there is such a slope, nearly constant with advance ratio, and varying little with incidence angle – remains to be analyzed.

Also plotted is the ratio power coefficient (C_P) vs. axial thrust coefficient (C_{TA}), for the range of incidence angles and advance ratios. Unless there is a Reynolds number effect, where propeller blade sectional performance varies with rotational speed because of increasing or decreasing Re , the “curve” of power coefficient vs. thrust coefficient will be a single point, if we do not vary the advance ratio or the incidence angle. This is another problem in the nondimensionalization of forces to coefficients in near-edgewise flights. If more vertical thrust is required, say, because the vertical component of the thrust of an inclined propeller is to be rescaled to the reference vertical thrust at $\theta = 90^\circ$, then, there should be a nonlinear increase in power required because the disk loading is now higher. But this increase is obviated if data are presented as coefficients, simply because the power coefficient is cubic in rpm, while the thrust coefficient is quadratic.

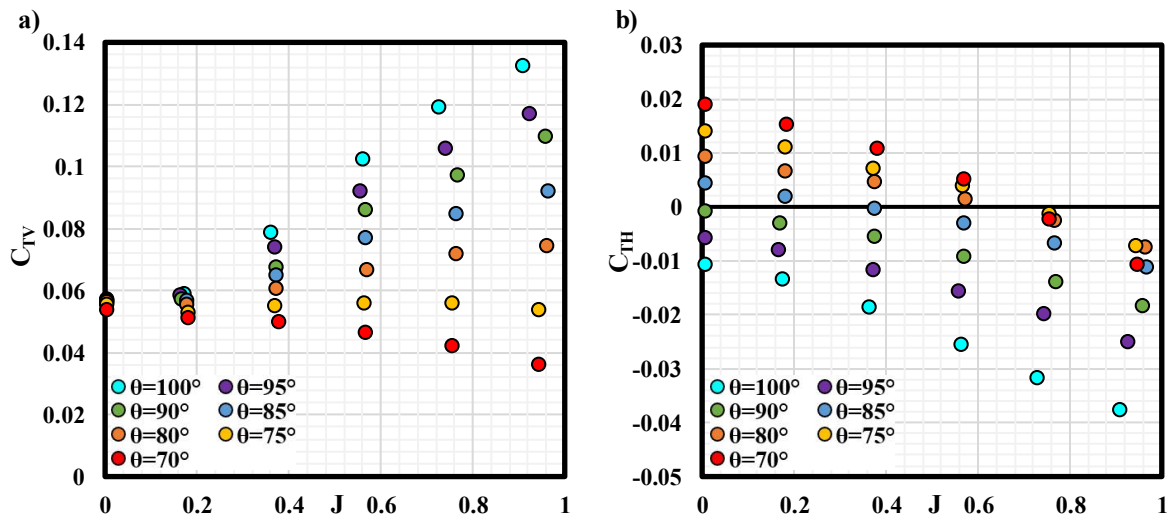


Figure 12. a) C_{TV} vs. J and b) C_{TH} vs. J for fixed fixed-pitch propeller at various θ .

The power coefficient for the fixed-pitch propeller is plotted in Figure 13a, again regarding only the shaft-power required to produce the axial thrust; that is, disregarding the in-plane force. The curves of C_P vs. J are more or less linear, with the most negative slope for $\theta = 100^\circ$, progressively reduced slope as θ is reduced until reaching $\theta \sim 70^\circ$, whereupon the trend reverses.

Results for C_P/C_{TA} curves vs. J are shown in Figure 14.a with a linear relationship, where the most negative slope for $\theta = 100^\circ$, nearly zero for $\theta \sim 80^\circ$, and positive for $\theta = 70^\circ$. Thus, if considering only the power-required to generate axial thrust in near-edgewise conditions, that is, mostly vertical thrust, the most efficient propeller orientation is slightly backwards-facing. Turning the propeller into the wing – that is, producing a useful component of thrust in the flight direction – incurs, in this presentation, a penalty in power-required. For near-edgewise flight, we wish to keep constant, or as a reference value, the vertical component of thrust.

Therefore, C_P/C_{TV} is also plotted in Figure 14 b. which represents the power consumption per unit of vertical force generated. The trends follow C_P/C_{TA} under all conditions, with a slightly different magnitude and slope.

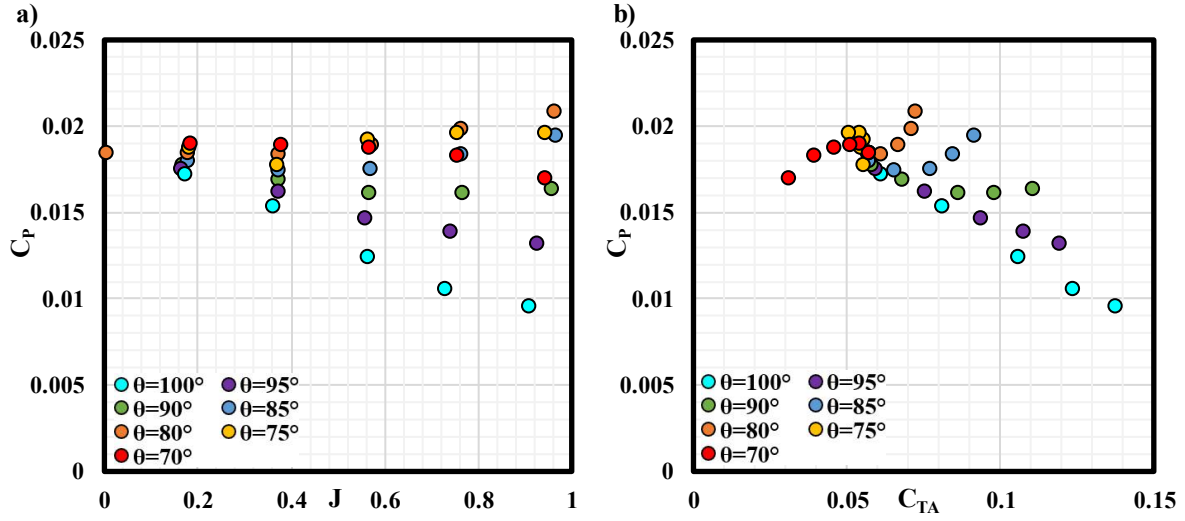


Figure 13. a) C_P vs. J and b) C_P vs. C_T for fixed fixed-pitch propeller at various θ .

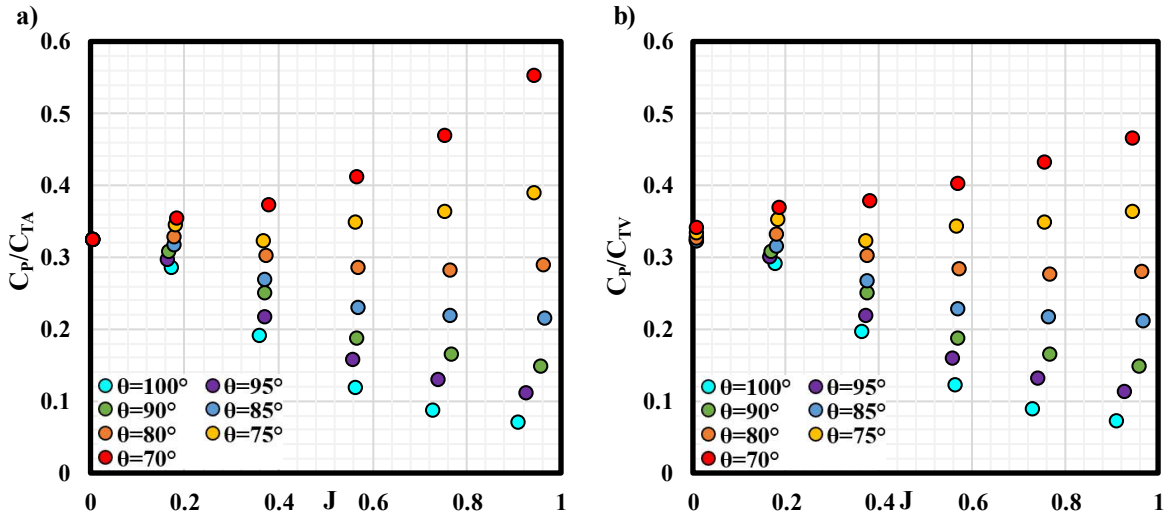


Figure 14. a) C_P/C_{TA} vs. J and b) C_P/C_{TV} vs. J for fixed fixed-pitch propeller at various θ .

C_P/C_{TA} or C_P/C_{TV} does not account for the penalty from the drag in the horizontal flight-direction in near-edgewise flight, which leads to a much lower C_P/C_{TV} ratio at larger θ . To account for the power deficit due to drag for $\theta \sim 90^\circ$, or the excess power at lower θ as a result of the forward thrust generated by the propeller (Figure 12), the total effective power coefficient is introduced, which accounts for the power consumption in both vertical and horizontal directions.

The first method in the current study uses the power consumption per unit total force generated by the propeller, that is, the vector summation of force on the propeller shaft and in-plane axes. One obtains C_P/C_{Total} at each θ and J . This ratio is then multiplied by C_{TV} and C_{TH} to obtain the effective propeller power coefficient for generating the vertical force (C_{PV}) and horizontal force (C_{PH}), respectfully. A positive C_{TH} represents forward thrust (Figure 12), and the resulting power is then excess power, taken as a credit or, more

precisely, a negative contribution to the overall power consumption. The sum of C_{PV} and C_{PH} is then the total power consumption coefficient ($C_{P_{Total}}$) of the propeller:

$$C_{P_{Total}} = C_{PV} + C_{PH} = C_{TV} \left(\frac{C_P}{C_{T_{Total}}} \right) + (-C_{TH}) \left(\frac{C_P}{C_{T_{Total}}} \right) \quad (2)$$

$C_{P_{Total}}$ vs. J using the first method is plotted in Figure 15 a. A nearly linear relationship is observed with a positive slope at $\theta < 80^\circ$ and a negative slope at $\theta > 80^\circ$. At lower J and near-hover conditions, a lower $C_{P_{Total}}$ is obtained at a lower θ , which is a result of the excess power due to the forward thrust generated by the propeller when tilted forward. At higher J , the differences between different θ first reduces; $C_{P_{Total}}$ merges at $J \sim 0.3$. Above this advance ratio, a lower $C_{P_{Total}}$ is observed for a larger θ , due to a much lower shaft power and the corresponding C_P magnitude measured seen in Figure 13. In other words, tilting the propeller forward, to produce a positive horizontal thrust,

An alternative method for $C_{P_{Total}}$ is to consider the additional power required or excess power from the propeller horizontal force (P_H) as the product of the propeller horizontal force (T_H) and the forward flight velocity (V_∞). A positive $T_H V_\infty$ is the excess power from the propeller's horizontal thrust, while a negative $T_H V_\infty$ is the additional power required to overcome drag. The total system power consumption (P_{Total}) is the sum of the propeller shaft power (Figure 13) and the excess/deficit power from the horizontal force. Based on this method, $C_{P_{Total}}$ can be written as

$$C_{P_{Total}} = C_P + C_{PH} = \frac{P_{shaft} + (-T_H)V_\infty}{\rho n^3 D^5} = C_P + (-C_{TH})J \quad (3)$$

$C_{P_{Total}}$ vs. J calculated from the second method is plotted in Figure 15 b. The overall trend resembles the power consumption vs. forward flight speed trend in traditional helicopter literature [13], that is, a reduction in $C_{P_{Total}}$ is observed with increase in J between $0 \leq J \leq 0.3$ at $\theta < 90^\circ$. After reaching its minimum, $C_{P_{Total}}$ then increases with J . At the same J , a lower $C_{P_{Total}}$ is obtained at a lower θ across the practical range of J for the fixed-pitch propeller.

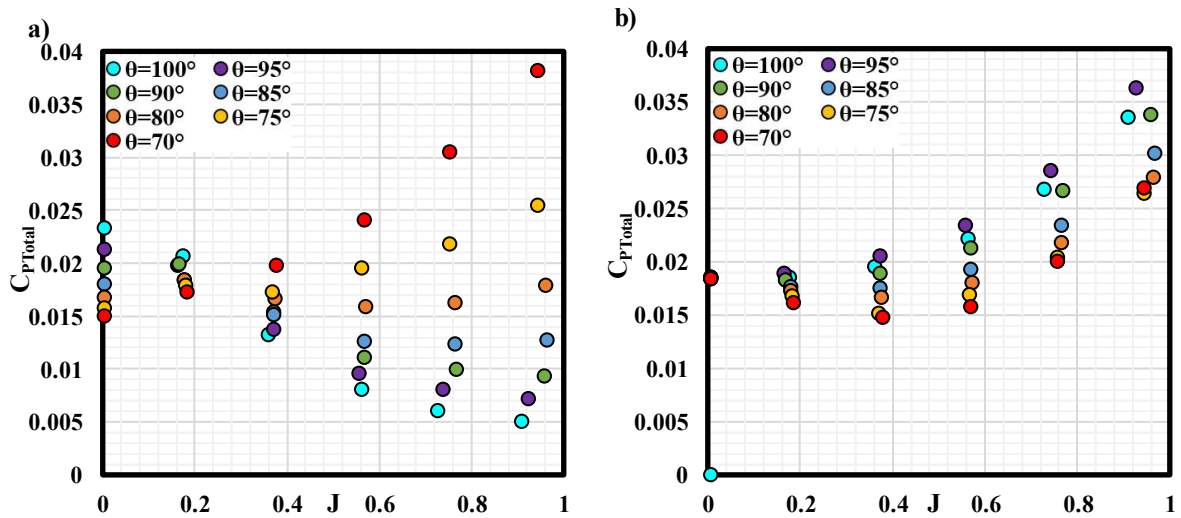


Figure 15. $C_{P_{Total}}$ vs. J for the fixed pitch propeller at various θ using a) Method 1 and b) Method 2.

Results in Figure 15b might suggest that a lower incidence angle in forward flight results in a lower overall power consumption. However, as discussed previously, the nondimensionalization process eliminates the effect of propeller rotational speed and the corresponding higher power consumption at a larger disk loading. For practical applications, the propeller will maintain a constant vertical thrust force in level forward flight. As seen in Figure 12, a lower flight incidence angle (θ) – that is, forward tilt - will result in a lower C_{TV} at the same forward flight speed, which would lead to a higher propeller rotational speed and thus, a larger disk loading to maintain a constant vertical thrust, resulting in a potentially larger overall power consumption. Therefore, we consider a constant vertical thrust force of 5N, which is the axial thrust generated by the fixed pitch propeller at the rotational speed tested. At each forward flight speed (V_∞) and incidence angle (θ), the propeller rotational speed is adjusted to meet such a vertical thrust requirement. The corresponding propeller shaft power (P_{shaft}) and the horizontal force (T_H) at the aforementioned flight condition is obtained. The resulting total power required (P_{Total}) at constant vertical thrust can then be calculated as:

$$P_{Total} = P_{shaft} + P_H = P_{shaft} + (-T_H)V_\infty \quad (4)$$

We close with a helicopter-style [13] plot of dimensional power (P_{total}) at constant dimensional vertical thrust ($T_V = 5N$), vs. dimensional flight speed, shown in Figure 16. Although these results are dimensional, they represent perhaps more clearly what happens to total power required, vs. flight speed, for various tilt angles. All tilt values nearly collapse to the same point in hover. At low forward speeds, the lowest forward tilt angle is best, because it provides a large forward thrust component (Figure 12 b), but incurs very little penalty in edgewise power-relief. By “edgewise power relief” we mean the reduction in power coefficient at constant thrust coefficient, considering only the vertical component. Then, as flight speed increases, this “edgewise power relief” becomes more prominent, and comes to overwhelm the forward thrust benefit of the highly tilted propeller. A propeller in purely vertical orientation looks to have the lowest total power at large forward flight speed. For drones, the implication evidently is, to tilt the lifting propeller – and hence the vehicle – forward, to rapidly accelerate from hover to cruise... but then to fly in edgewise flight, using a “helper” axial propeller to provide the horizontal thrust to overcome total system drag. However, the trends shown in Figure 16, except for the lowest forward tilt angle, are not large in relative magnitude. There is no obvious optimum. Designers have a broad choice, or perhaps, the results have broad ambiguity.

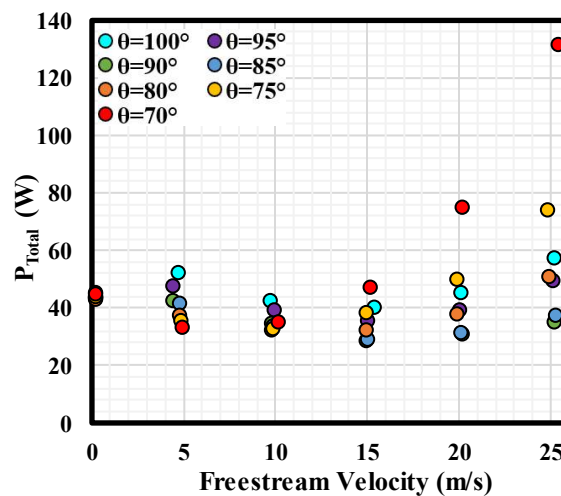


Figure 16 Power required at constant vertical thrust (5N) vs. freestream velocity for fixed pitch propeller.

V. Conclusions and Future Work

Increasing the collective pitch of a single propeller in nearly-edgewise flight – that is, with its thrust axis some 10 degrees backwards to 30 degrees forwards, with respect to the vertical – does not show benefit in efficiency. The implication is that variable collective pitch for a VTOL multi-copter drone, for all of its potential benefits in flight control, does not offer a benefit in propulsive efficiency. Next comes the question of how to reckon with the in-plane force, or effectively the drag of a propeller in nearly-edgewise flight. This can be addressed by tilting the propeller forward, where a projection of its [axial] thrust compensates for its drag; or the drag can be compensated by other means, where these means have to be accounted for, in terms of additional power required. It was found that the optimum is weak; purely edgewise flight, or tilting forward by 5-10 degrees, results in essentially the same overall power budget, potentially giving drone designers more freedom to choose configurations to suit. However, the ambiguity of how to nondimensionalize forces in edgewise (as opposed to axial) flow is a question to be posed to the community: how best to present our results, and how to rigorously account for the role of propeller in-plane force (drag)?

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